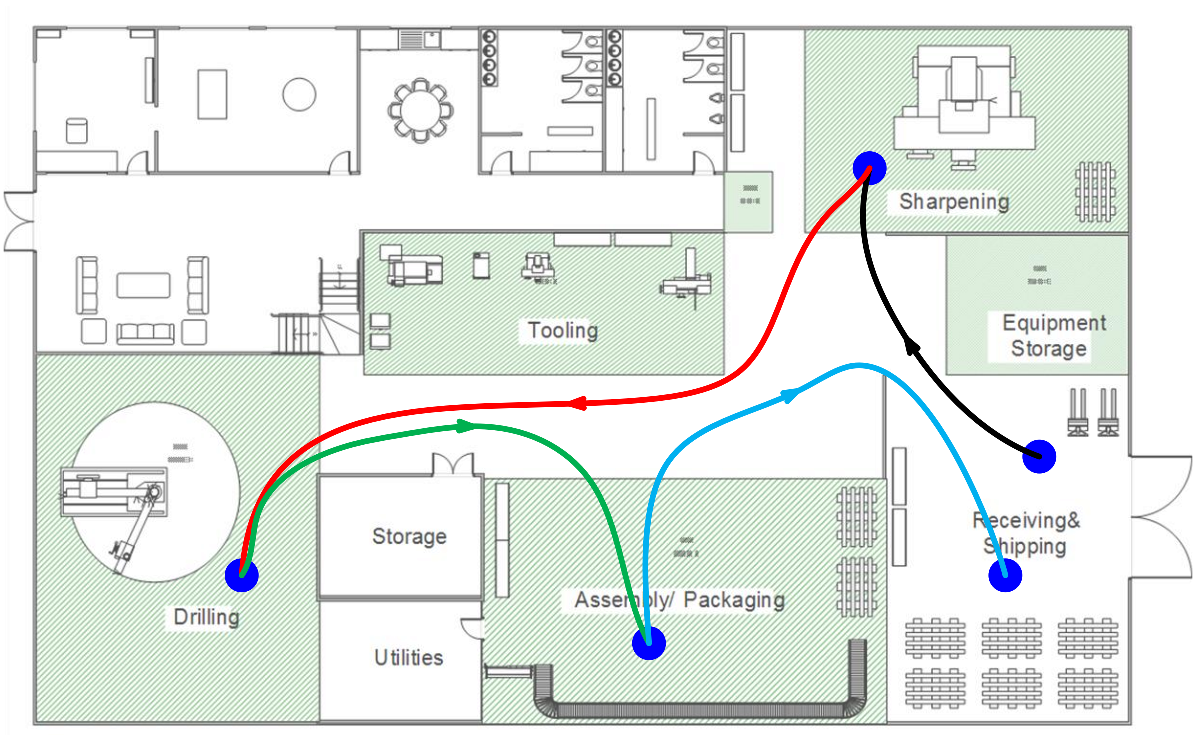
Spaghetti Diagrams

Spaghetti diagramming is a visual tool used in process improvement and optimization. This course shows the eVSM spaghetti diagramming capability and provides step-by-step instructions to use it.



eVSM Version: v12

Course version: 003

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www.evsm.com

How to Use this File

This file contains the reading materials and the exercise pages from the course (title on previous page). While the course can only be taken on a computer, this booklet can be useful for note taking and later for refresher training.

This booklet is designed for on screen and print use. For on screen use, we recommend Acrobat Reader with the page display set to "Single Page View". If you are using this booklet on-screen while going through the exercises in eVSM, a second monitor is very helpful.

For hardcopy use, print the file on 8.5x11 or A4, and bind along the long edge.

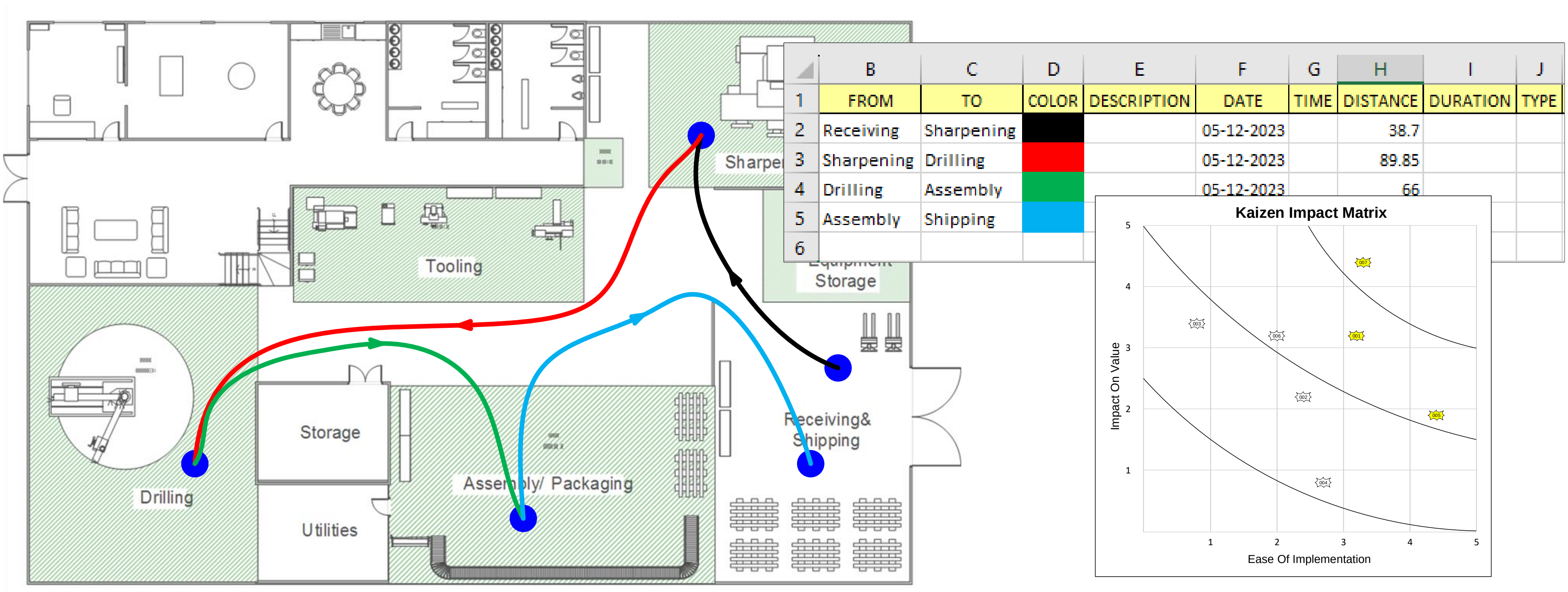
Table of Contents

Spaghetti Diagram

Spaghetti diagramming is a visual tool used in process improvement and optimization. It helps identify and analyze the flow of people, materials, or information within a physical space. Spaghetti diagrams can give insights into the inefficiencies, bottlenecks, or waste within a process or layout. By mapping out the actual paths, it becomes easier to identify unnecessary movements, excessive distances traveled, or potential areas for improvement.

In this course, you will learn how to create spaghetti diagrams in eVSM.

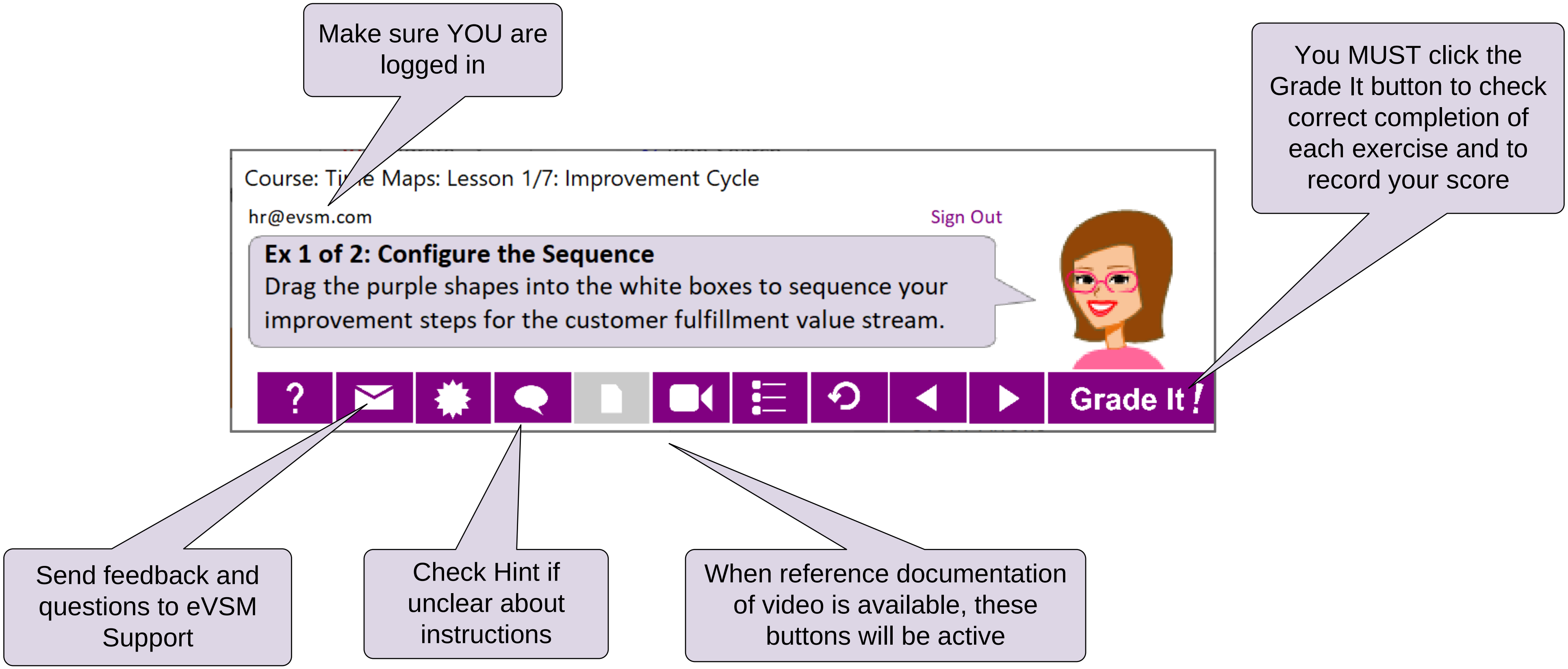
eVSM Spaghetti Diagram Course



You must have eVSM v12.22 or later to take this course.

Working with eLeanor

The eLeanor learning system includes a range of useful functions:



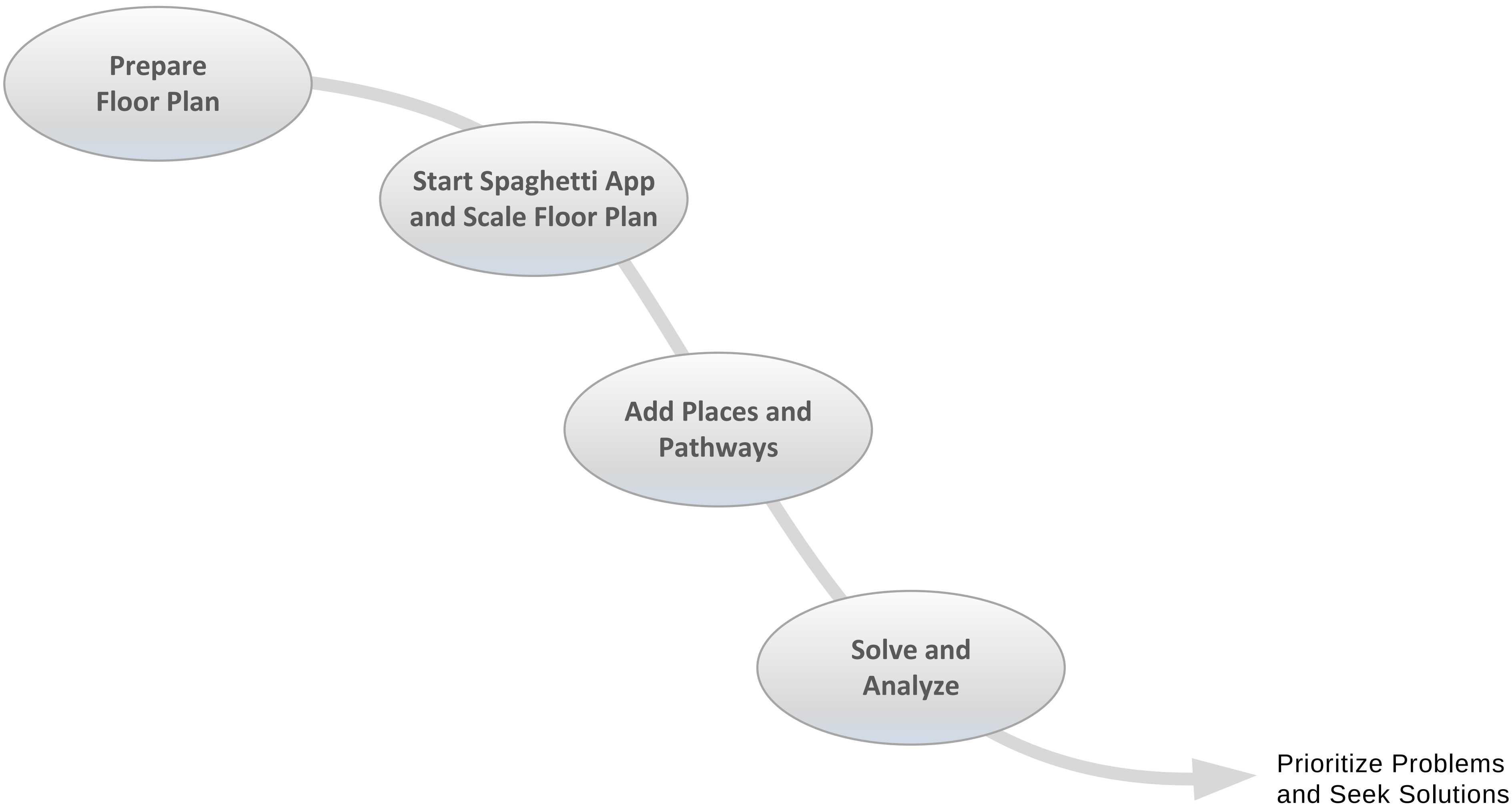
Important Notes

1. When you complete an exercise, you **MUST** click the “Grade It” button.
2. Points are deducted for incorrect attempts.
3. If you are stuck on an exercise, check the Hint. If that does not help, go back and review the preceding Readme pages. If you are still unsure, click the Feedback button and ask your question.

eVSM Spaghetti Diagram Process

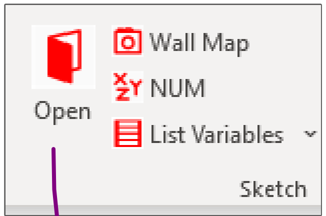
The eVSM Spaghetti Diagramming application allows you to sketch transport pathways on a floor plan and then generate an Excel report showing journey segment distances and related data.

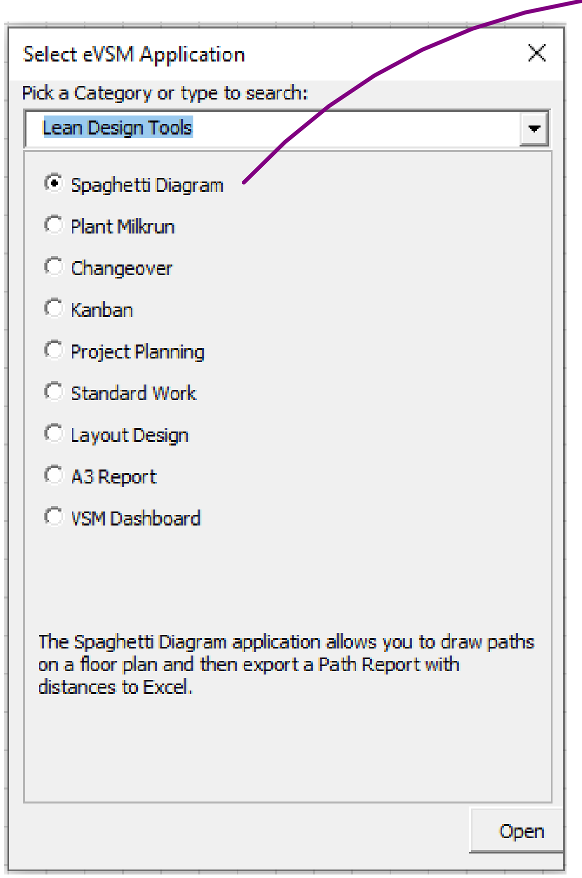
Here is an outline of the steps:

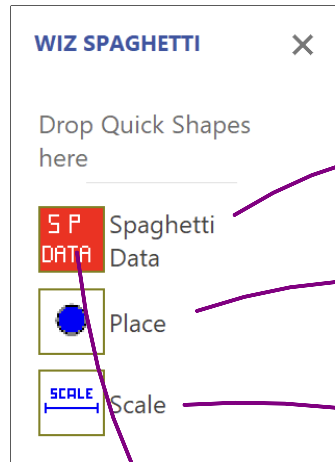


Getting Started with Spaghetti Diagram

1 Open the Spaghetti Diagram application







Used to initiate the page for the spaghetti diagramming application.

The Place shape is used to represent the activity places on the floor plan.

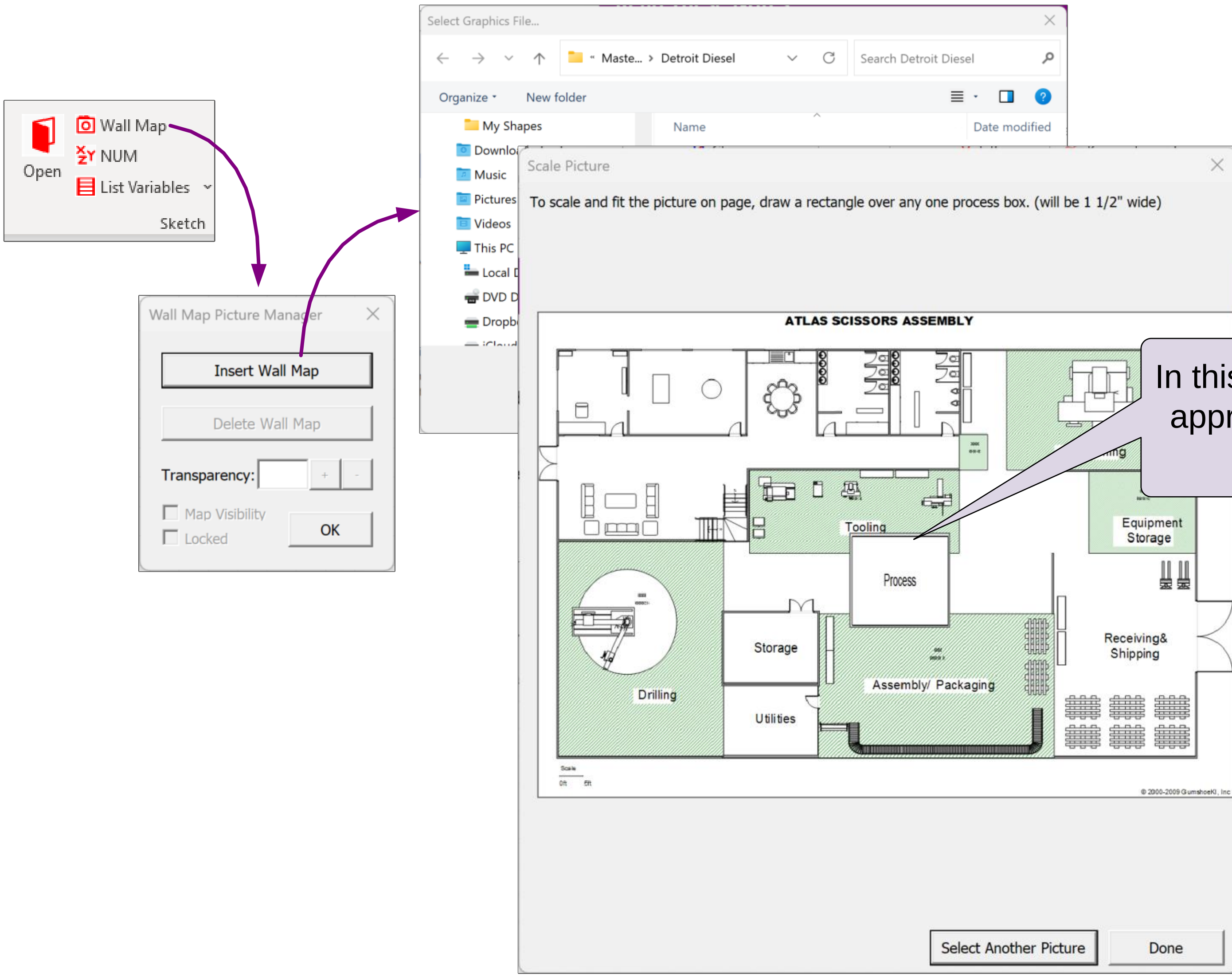
Scale shape is required to specify the scale for the floor plan.

2 Initiate a spaghetti diagram by dropping the Spaghetti Data shape anywhere on the page.

The Floor Plan

The floor plan may be drawn in Visio, or imported in bitmap or line drawing format. It can be a photo of the fire escape diagram, screenshot of a Google map, etc.

There are several ways of bringing an external image onto a Visio page. We recommend using the Wall Map button in the eVSM toolbar. Here are the steps:



In this preview pane, draw a square approximately 1"x1" (2.5 cm x 2.5 cm) and then click Done.

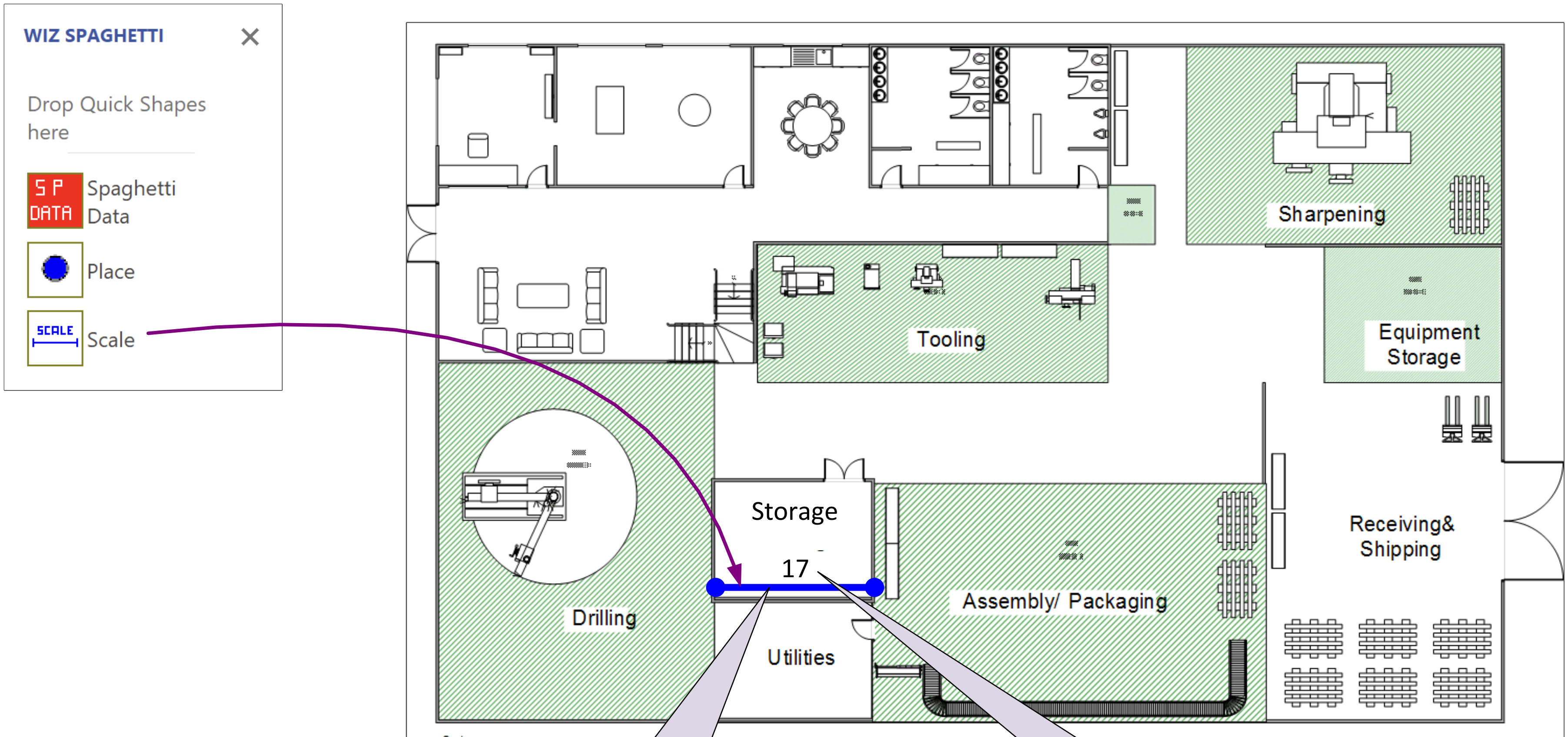
This will automatically size the page to fit the image, and it will lock the image on the page so it does not get nudged when you draw the spaghetti diagram.

Initiate Spaghetti Diagram and Import Floor Plan

1. Initiate this page for an eVSM spaghetti diagram.
2. Use the “Wall Map” button to import the floor plan in the **MFGfloor.gif** (this is included in the file set you downloaded for this course).

Specify the Floor Plan Scale

The Scale shape is used to specify the scale of the floor plan. Once the floor plan image is in place, drop one scale shape on the page and stretch it out to line up with a known dimension on the page such as length of a room, width of a door, etc. Note that you must have only one scale shape on the page. Also, the scale shape has no units. This tool does not actually know about units, so you may be working in whatever units you wish.

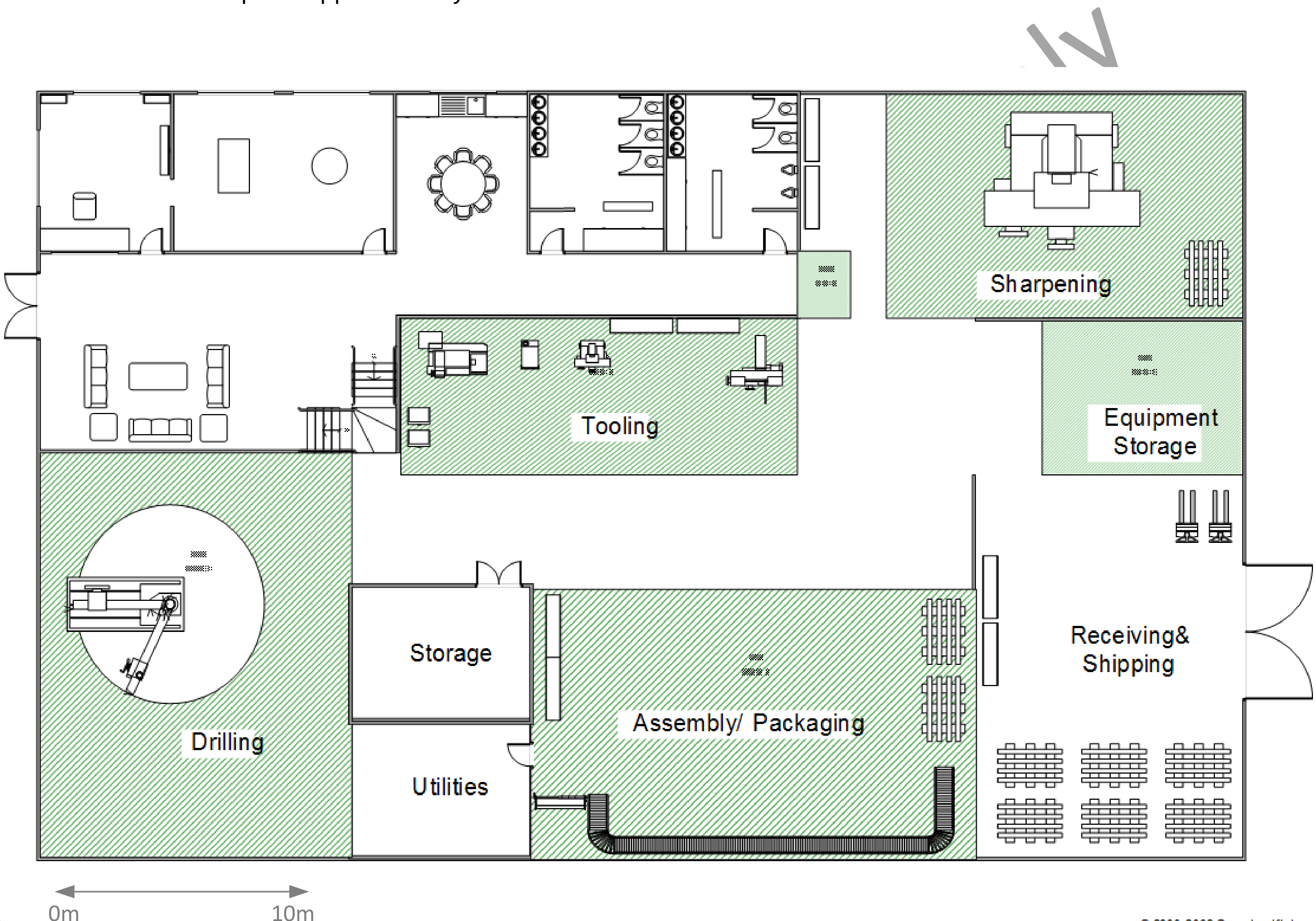


Stretch out the scale shape to align with the known dimension on the image.

Enter the measured dimension. DO NOT specify any units.

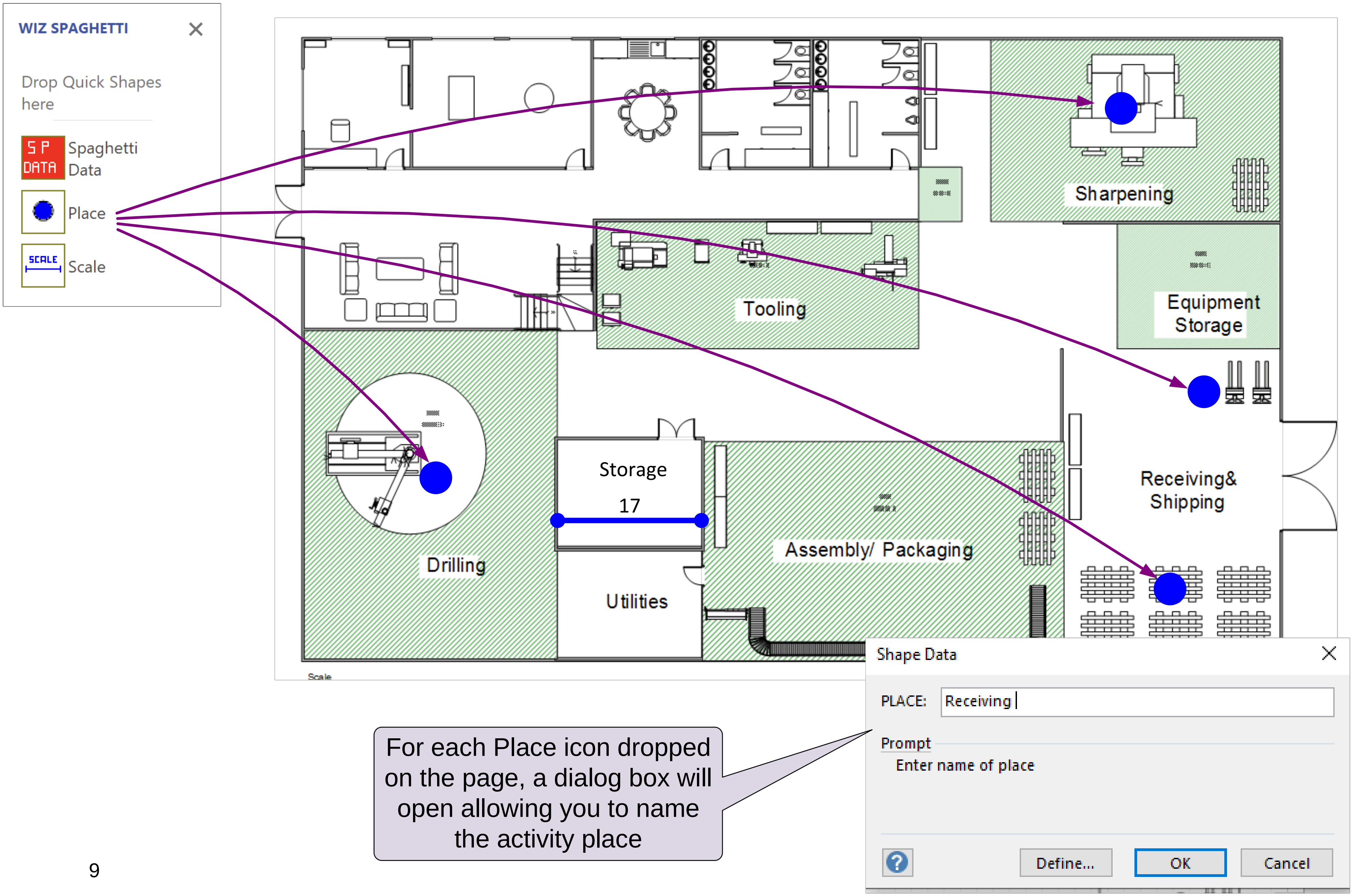
Spaghetti Scale Exercise

1. Initiate this page for a Spaghetti diagram.
2. Drop a "Scale" shape on the 10m arrow near the bottom left corner of the floorplan image.
3. Stretch out the Scale shape to approximately fit the arrow and enter "10". DO NOT enter "10m".



Specify Activity Places on the Floor Plan

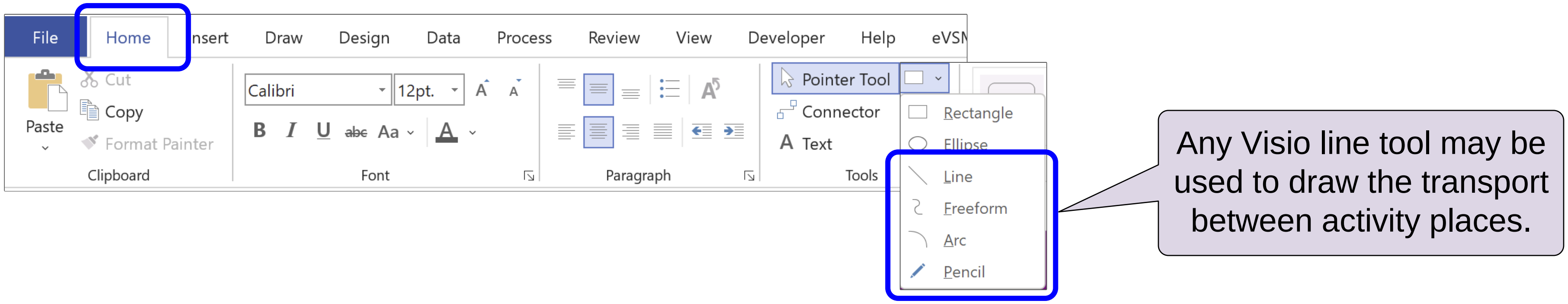
The Place shape is used to represent the activity places involved in the process.

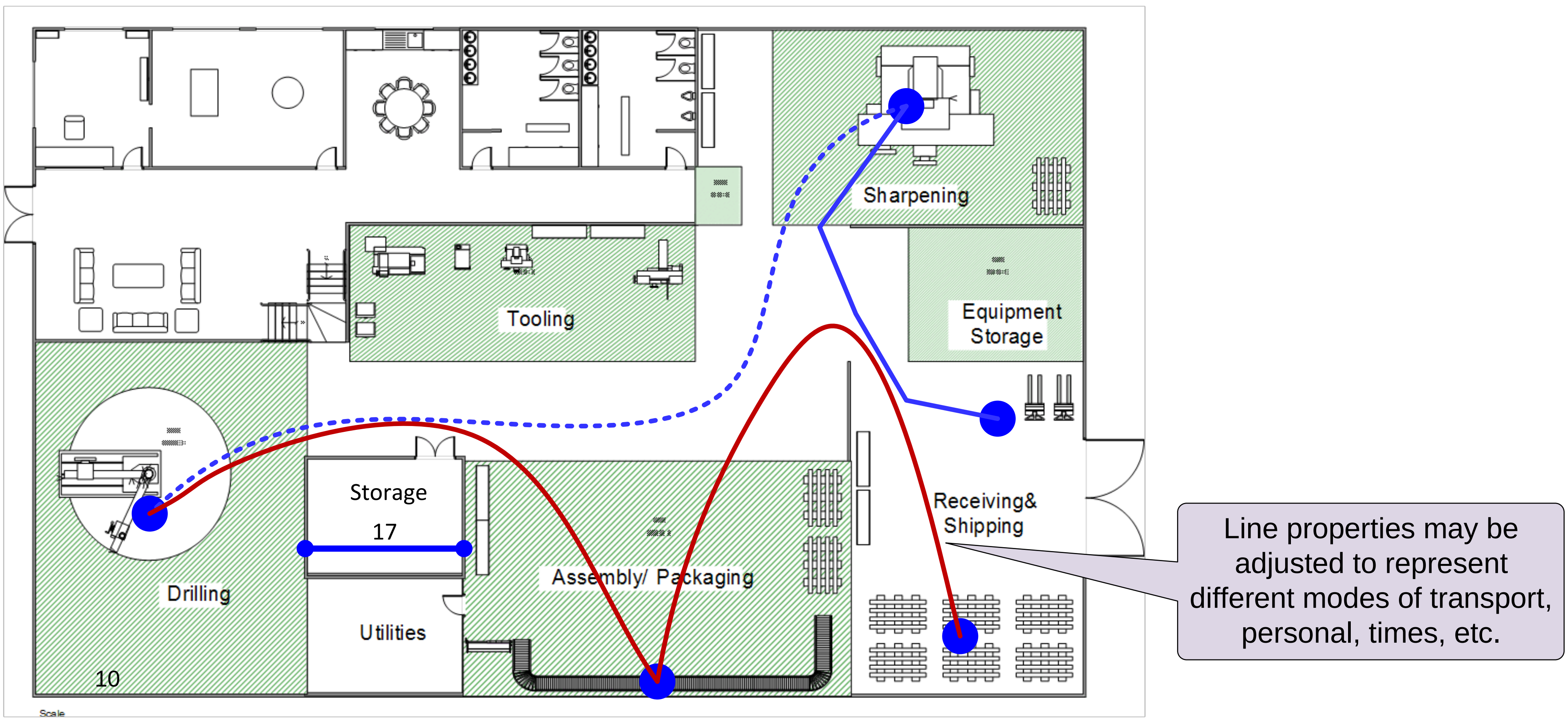


For each Place icon dropped on the page, a dialog box will open allowing you to name the activity place

Draw the Transport Paths

Draw the transport paths between the place icons with ANY Visio line drawing tool. Each spaghetti path line must begin and end on a Place icon. Lines that do not meet this requirement will be ignored by the analyses.

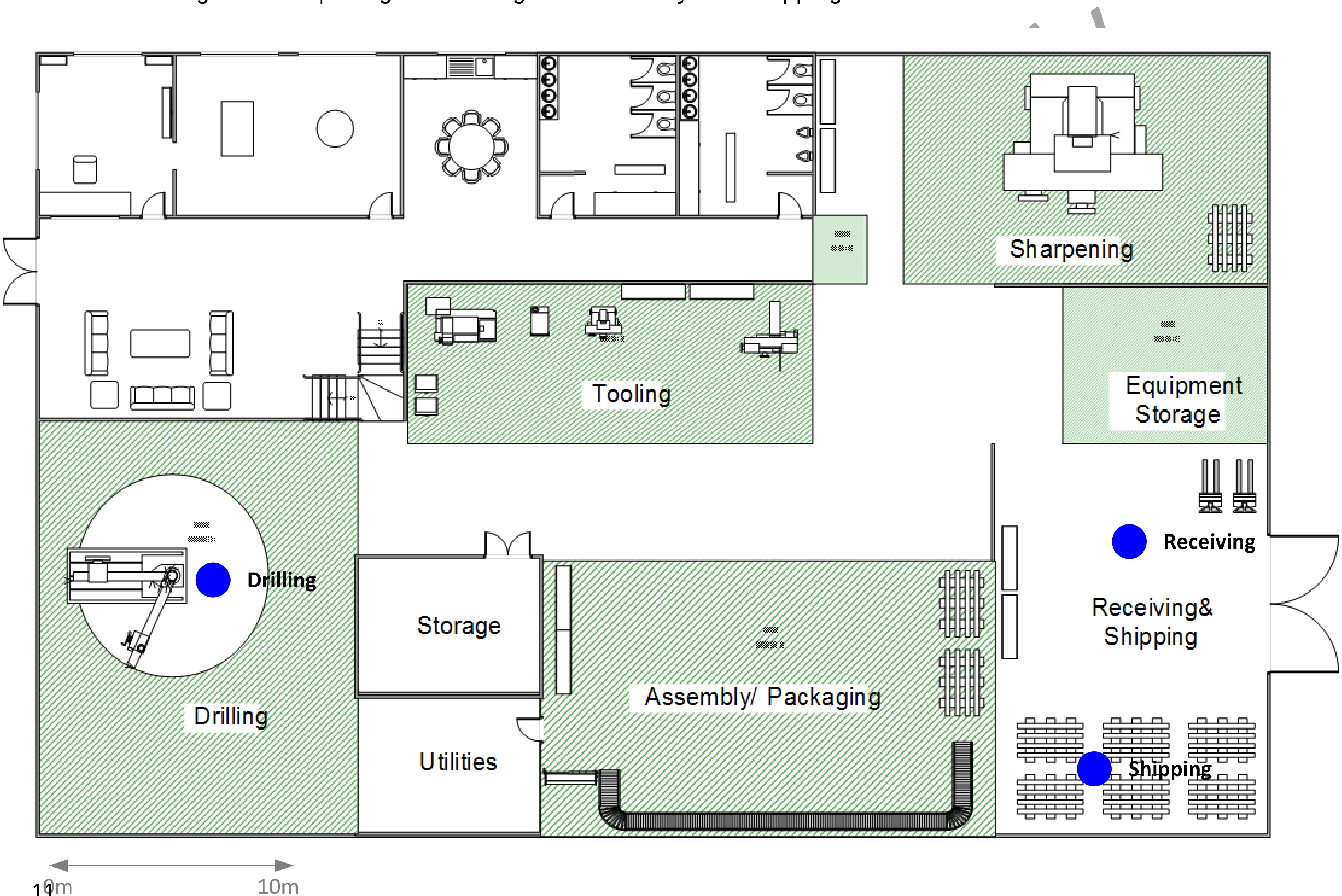




Spaghetti Places and Paths Exercise

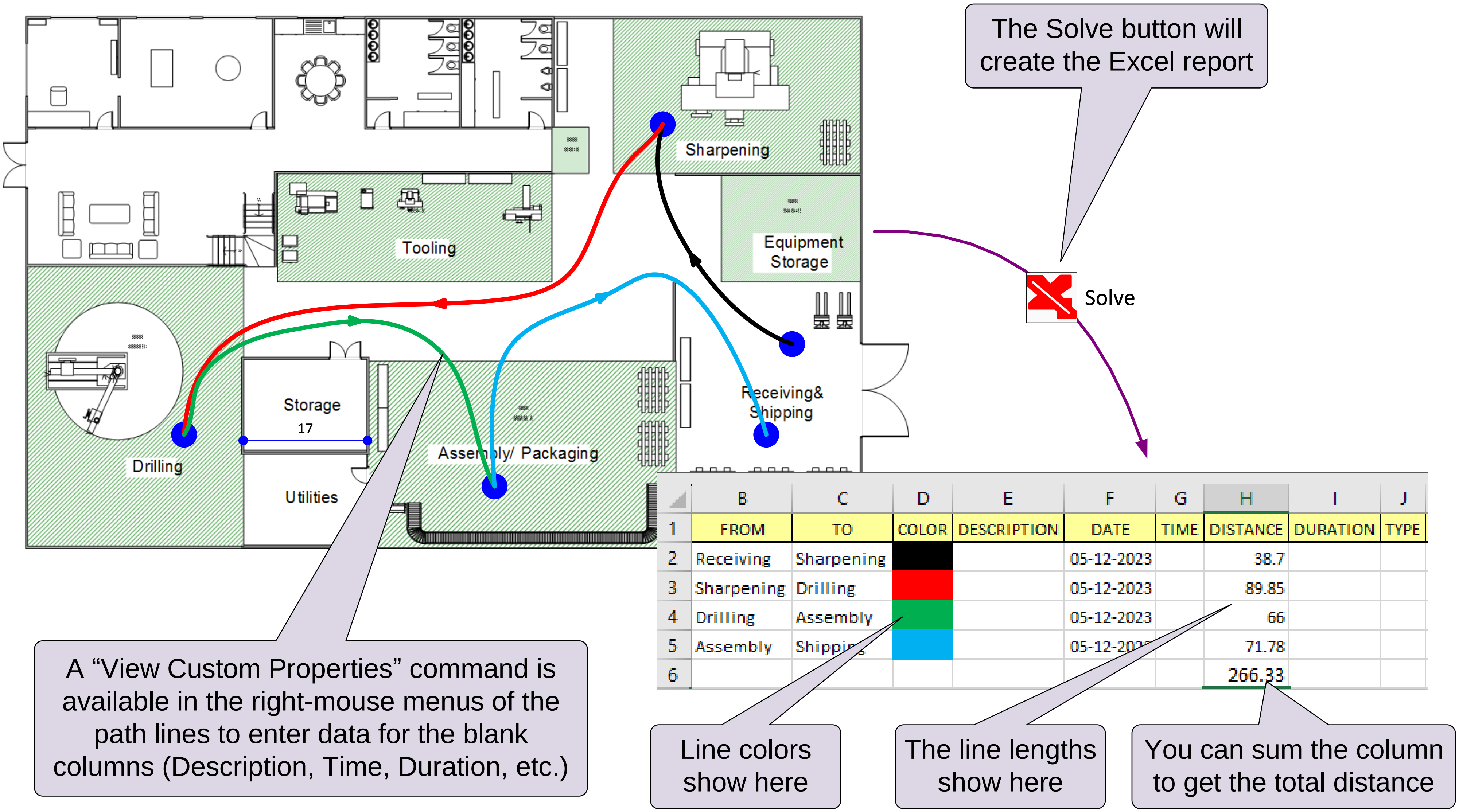
1. Add Spaghetti Place shapes to the Sharpening and Assembly areas.
2. Using the Visio line drawing tools, draw these transport paths:

Receiving → Sharpening → Drilling → Assembly → Shipping



Solve and Analysis

Once the Scale, the Places, and the transport paths are established on the page, click the Solve button to generate and Excel report.



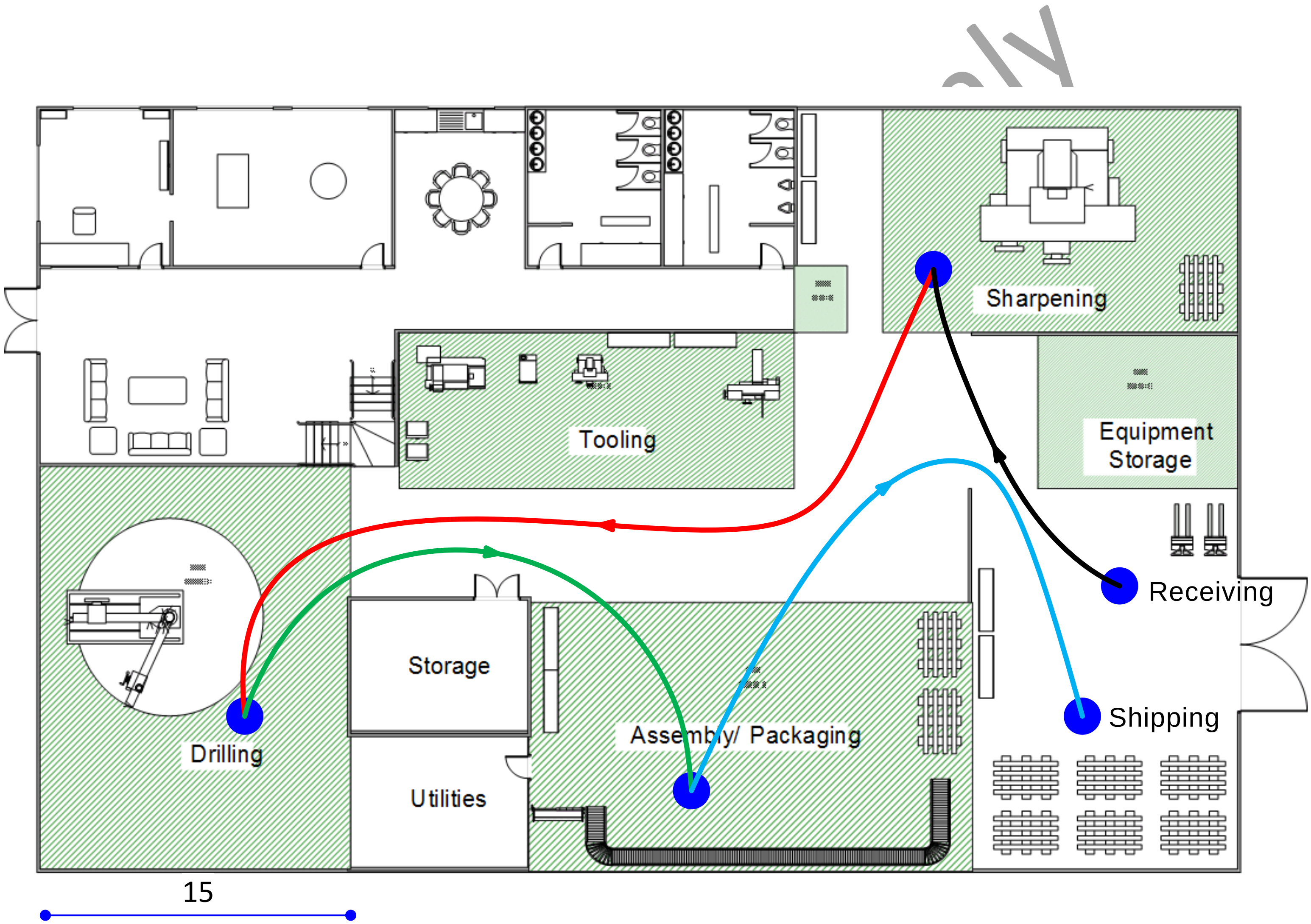
	B	C	D	E	F	G	H	I	J
1	FROM	TO	COLOR	DESCRIPTION	DATE	TIME	DISTANCE	DURATION	TYPE
2	Receiving	Sharpening			05-12-2023		38.7		
3	Sharpening	Drilling			05-12-2023		89.85		
4	Drilling	Assembly			05-12-2023		66		
5	Assembly	Shipping			05-12-2023		71.78		
6							266.33		

IMPORTANT: The data flows one-way from Visio to Excel. There is no return trip. Changes made in Excel DO NOT come back to Visio. Also, the next time the Solve button is invoked, the Excel file will be overwritten.

What is the total material movement travelled from Receiving to final Shipping?

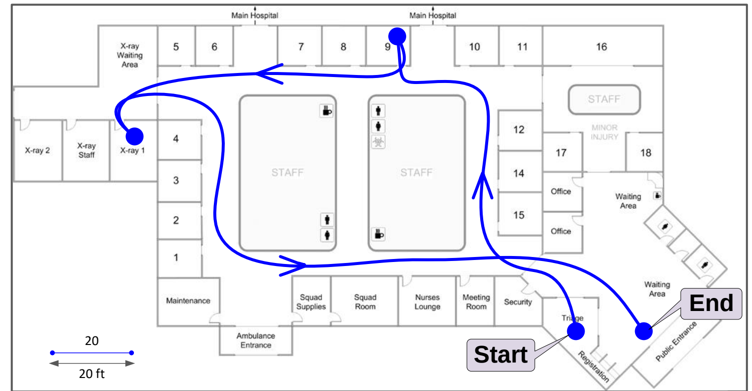
Solve the model and calculate the total in the Excel report.

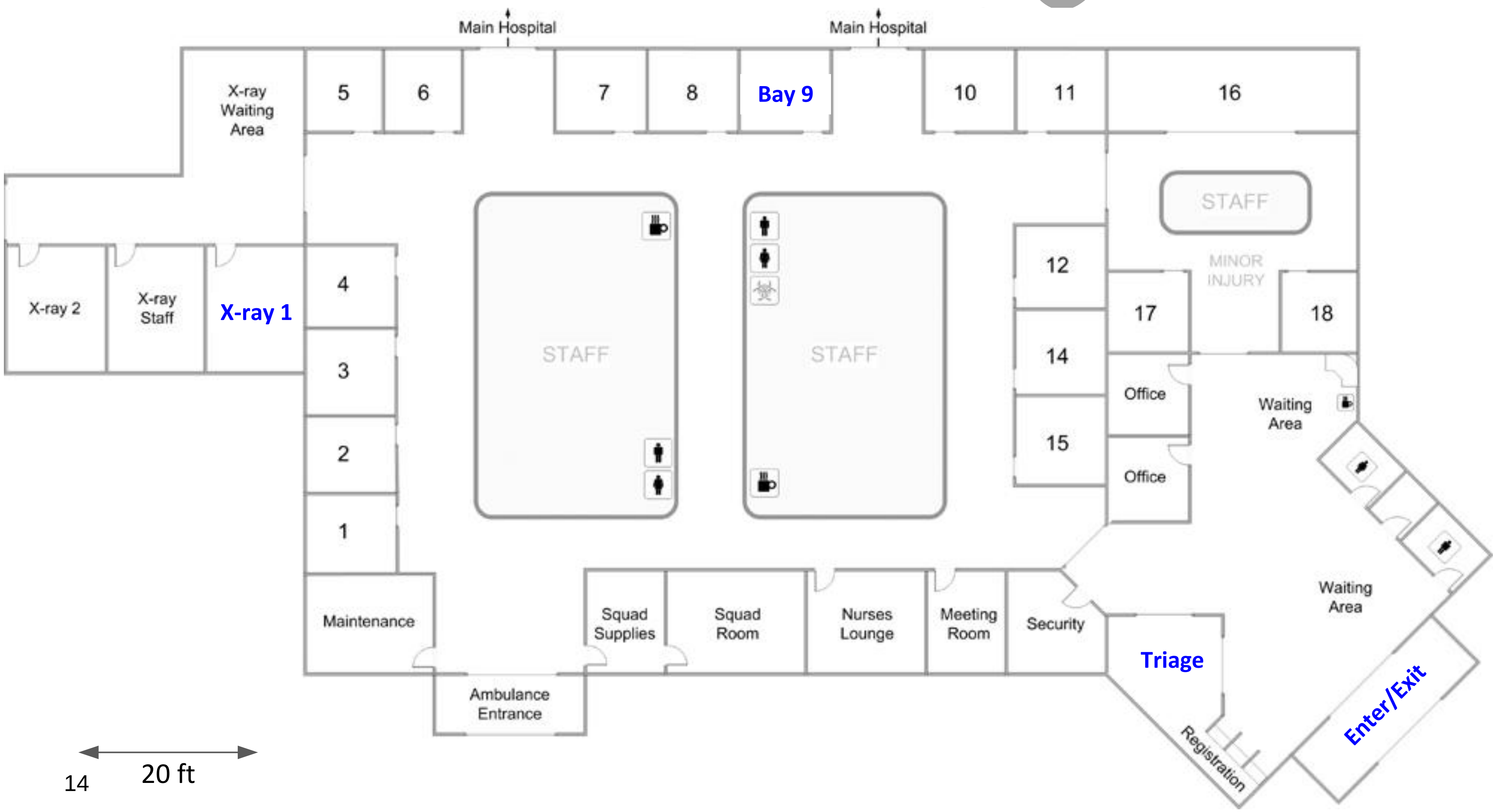
- ◎ 142 units
- ◎ 135 units
- ◎ 139 units
- ◎ 148 units



Patient Flow Spaghetti Diagram

1. Open the Spaghetti application and Initiate this page for a Spaghetti diagram
2. Use the “Scale” shape to set the scale
3. Add Spaghetti places and draw the paths as shown in the image on the right (Triage > Bay 9 > X-ray 1 > Exit)
4. Solve the model and enter the total patient travel distance here [ ] ft

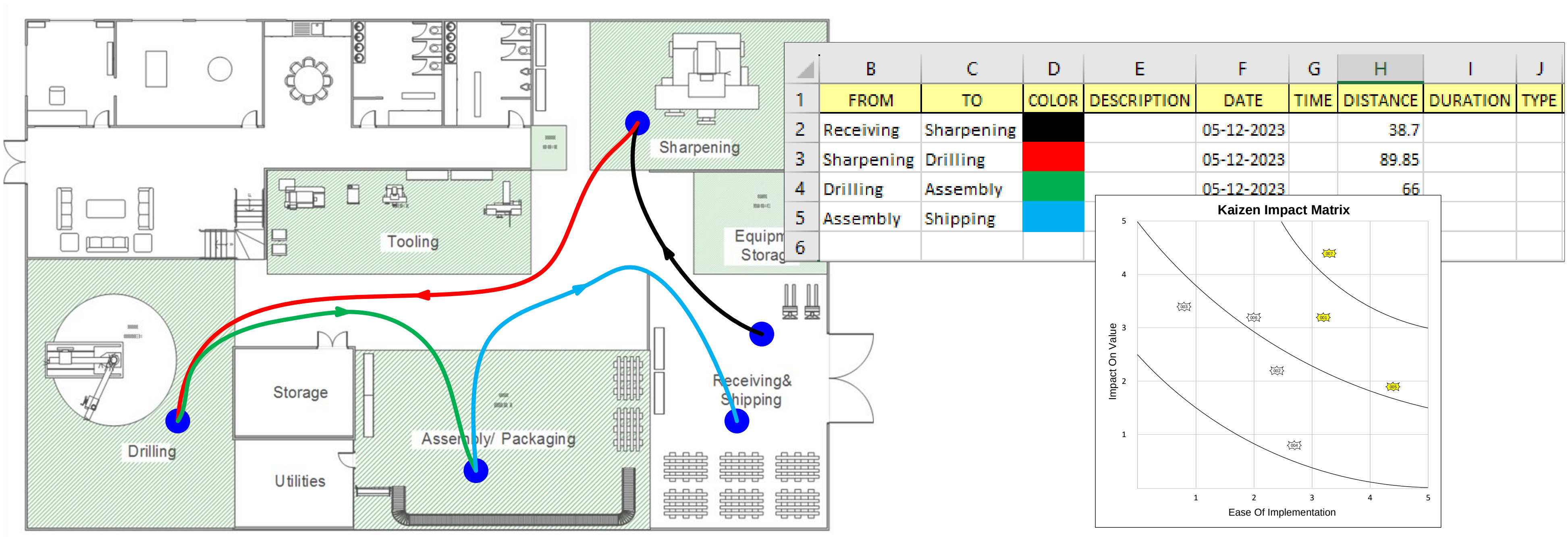




You Learned That :

- The eVSM Spaghetti Diagramming tool can be used to show transport on a floor plan. & Calculate the total distances and times
- The steps to create a Spaghetti diagram model in eVSM
- What-if analyses for proposed changes to the layout

eVSM Spaghetti Diagram Course



	B	C	D	E	F	G	H	I	J
1	FROM	TO	COLOR	DESCRIPTION	DATE	TIME	DISTANCE	DURATION	TYPE
2	Receiving	Sharpening			05-12-2023		38.7		
3	Sharpening	Drilling			05-12-2023		89.85		
4	Drilling	Assembly			05-12-2023		66		
5	Assembly	Shipping							
6									

Next Step:

- Try this on your floor plan
- Review/Learn the Facility Layout Design tool in eVSM

Useful Links

eVSM Toolbar Guide	evsm.com/toolbarguide
eVSM Productivity Guide	evsm.com/productivity
eVSM Blogs	evsm.com/blog
eVSM Support FAQ	evsm.com/support
Download the Latest Version	evsm.com/install